

A220 · Structural Modeling Identifies a Putative, MIF-Independent CD74–IFNGR1 Interface in IFN- γ Signaling

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CD74 deletion abolishes IFN- γ -driven allograft rejection (median survival >100 days vs. 7 days) and reduces T-cell pSTAT1 signaling by ~50% despite unchanged surface IFNGR1, indicating a receptor-proximal defect that persists in MIF-deficient mice, demonstrating that the canonical CD74 ligand is dispensable. To identify a structural mechanism, we integrated sequence analysis, experimentally resolved receptor structures, multimeric modeling, multi-platform protein–protein docking, interface energetics, and molecular dynamics. IFNGR1–CD74 sequence comparison revealed 24.0% global and 29.5% local identity over a 78-residue window. Mapping this region onto IFN- γ receptor complexes 6E3K and 6E3L identified an extracellular IFNGR1 segment (~129–221; predominantly 162–221) overlapping the experimentally defined IFN- γ /IFNGR1 interface at 25/75 and 24/72 resolved residues, respectively, with approximately half of these residues solvent-exposed. Independent HDock, ClusPro, and HADDOCK docking converged on this region, supporting a reproducible topology involving multiple CD74 protomers. HDock favored CD74 alone over CD74–MIF (–348.32 vs. –275.48), arguing against an MIF requirement for the predicted association. Interface energetics identified 16 dominant CD74–MIF hotspots ($\Delta G \leq -10$ kcal/mol), with stabilization correlated with solvent burial ($R^2=0.79$, $p<0.01$). Complementary 200-ns MD supported interface persistence: backbone RMSD remained below 3 Å, while 68% of candidate interface residues maintained contacts for >50% of the trajectory. Together, these orthogonal analyses identify a structurally plausible, multivalent CD74–IFNGR1 association overlapping the physiological IFN- γ -binding interface, providing a structural correlate for the MIF-independent receptor-proximal defect and nominating residues for experimental validation.